

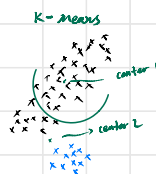
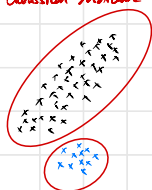
Gaussian mixture model is a type of clustering model that assumes the instances are generated from a mixture of Gaussian Distributions.

Compare with K-means

Instances:



Gaussian Mixture model



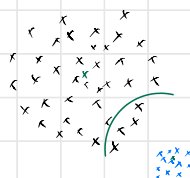
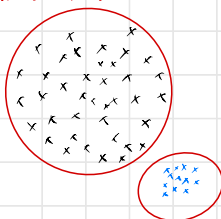
K-means Cannot handle elliptical clusters well

because it creates circles or balls around its centers

Instances:



Gaussian Mixture Model



K-means Cannot deal with clusters with different variance well

Because it uses the same distance measure (normally L_2) for different clusters

Assumptions of Gaussian Mixture model

→ Likelihood of observations

$$pr(x_1, \dots, x_n) = pr(x_1) \dots pr(x_n)$$

independent assumption

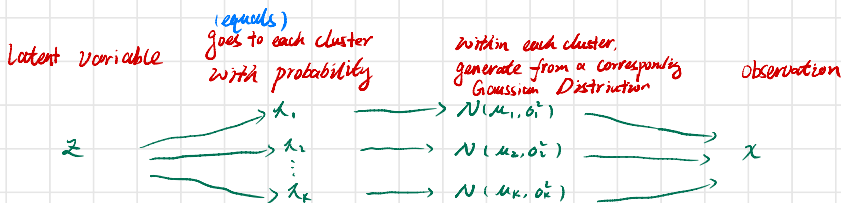
$$= \prod_{i=1}^n pr(x_i \in \text{cluster } j) pr(x_i | x_i \in \text{cluster } j)$$

Law of total probability

$$= \prod_{i=1}^n \sum_{j=1}^K \pi_j \mathcal{N}(x_i; \mu_j, \sigma_j^2)$$

Gaussian Assumption

Generation Process:



The latent variable z is a discrete variable equal i (where $i=1, \dots, K$) with probability π_i . Given this latent variable, the likelihood of the data x is normally distributed with mean $= \mu_i$, var $= \sigma_i^2$.

It is hard to maximize the likelihood function

$$pr(x_1, \dots, x_n) = \prod_{i=1}^n \sum_{j=1}^K \lambda_j \mathcal{N}(x_i; \mu_j, \sigma_j^2)$$

directly. $\downarrow$ minimize negative log-likelihood

$$\begin{aligned} \min_{\lambda, \mu, \sigma} (-L(\lambda, \mu, \sigma; x_1, \dots, x_n)) &= \min_{\lambda, \mu, \sigma} - \sum_{i=1}^n \log \left(\sum_{j=1}^K \lambda_j \mathcal{N}(x_i; \mu_j, \sigma_j^2) \right) \\ &= \min_{\lambda, \mu, \sigma} - \sum_{i=1}^n \log \left(\sum_{j=1}^K \lambda_j \frac{1}{(\sigma_j^2)^{\frac{1}{2}} \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} (x_i - \mu_j)^T \Sigma_j^{-1} (x_i - \mu_j) \right\} \right) \end{aligned}$$

$$\text{s.t. } \sum_{j=1}^K \lambda_j = 1$$

$\swarrow$ Σ_j is symmetric and positive semi-definite

$\checkmark$ because of this sum, we cannot use log to cancel out exp there is no analytical form if we take gradient w.r.t λ, μ, σ and set the gradients to 0

$\downarrow$ need to satisfy the constraints,

so directly use gradient descent is not ideal

★ The problem is also ill-defined. If $\mu_j \rightarrow x_i$, $\Sigma_j \rightarrow 0$, then $pr(x_1, \dots, x_n) \rightarrow \infty$

Below we consider how to use E-M algorithm to solve these issues.

E-M algorithm

problem: $\arg\max_{\theta} L(\theta; x) \rightarrow$ this is hard

Observation: If we introduce some latent variables z , then

$\arg\max_{\theta} L(\theta; x, z) \rightarrow$ this is easy

E-M algorithm:

$$\theta^{(t+1)} = \arg\max_{\theta} \underbrace{\bar{L}(\theta; x, z)}_{\text{M-step}} \quad \underbrace{z \sim p(\cdot | x, \theta^{(t)})}_{\text{E-step}}$$

Why does E-M algorithm work?

Notice that

$$p(x, z; \theta) = p(x; \theta) p(z|x; \theta)$$

$$\Rightarrow \ell(\theta; x, z) = \ell(\theta; x) + \ell(\theta; z|x)$$

$$\Rightarrow \underbrace{\ell(\theta; x)}_{\text{target}} = \underbrace{\ell(\theta; x, z)}_{\text{the one we can maximize easily}} - \ell(\theta; z|x)$$

$$\Rightarrow \ell(\theta; x) = \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; z|x)) \quad \text{take integration on both sides}$$

$$\Rightarrow \ell(\theta^{(t)}; x) = \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z)) - \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; z|x))$$

$$\begin{aligned} \Rightarrow \ell(\theta; x) - \ell(\theta^{(t)}; x) &= \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z)) \\ &\quad - \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; z|x)) + \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; z|x)) \\ &= \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z)) \\ &\quad + \underbrace{\text{KL}(p(z|x, \theta^{(t)}), p(z|x, \theta))}_{\rightarrow \geq 0} \end{aligned}$$

Since

$$\theta^{(t+1)} = \underset{\theta}{\operatorname{argmax}} \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z))$$

We have

$$\mathbb{E}_{z \sim p(\cdot|x, \theta^{(t+1)})} (\ell(\theta^{(t+1)}; x, z)) \geq \mathbb{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t+1)}; x, z))$$

So $\ell(\theta^{(t+1)}; x) \geq \ell(\theta^{(t)}; x)$, i.e. E-M algorithm leads to non-decreasing likelihood function at each step.

E-M algorithm for Gaussian Mixture model

Introduce latent variables: $z_{ij} = \begin{cases} 1 & \text{if } x_i \in C_j \\ 0 & \text{otherwise} \end{cases}$

In order to use E-M algorithm, we need joint distribution $p(x, z; \theta^{(t)})$ and conditional distribution $p(z | x; \theta^{(t)})$.

E-Step:

Joint distribution

$$p(x, z; \kappa, \mu, \Sigma) = \prod_{i=1}^n \prod_{j=1}^K \left(\kappa_j \mathcal{N}(x_i; \mu_j, \Sigma_j) \right)^{z_{ij}}$$

$$\Rightarrow L(\kappa, \mu, \Sigma; x, z) = \sum_{i=1}^n \sum_{j=1}^K (z_{ij} (\log \kappa_j + \log \mathcal{N}(x_i; \mu_j, \Sigma_j)))$$

Conditional distribution

$$p(z_{ij}=1 | x; \kappa^{(t)}, \mu^{(t)}, \Sigma^{(t)}) = p(z_{ij}=1 | x_i, \kappa^{(t)}, \mu^{(t)}, \Sigma^{(t)})$$

$$= \frac{p(z_{ij}=1; \kappa^{(t)}) p(x_i | z_{ij}=1, \kappa^{(t)}, \mu^{(t)}, \Sigma^{(t)})}{\sum_{j=1}^K p(z_{ij}=1; \kappa^{(t)}) p(x_i | z_{ij}=1, \kappa^{(t)}, \mu^{(t)}, \Sigma^{(t)})}$$

Bayes Theorem

$$= \frac{\kappa_j^{(t)} \mathcal{N}(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}{\sum_{j=1}^K \kappa_j^{(t)} \mathcal{N}(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}$$

$$= p_{ij}^{(t)}$$

So we have

$$\bar{E}_{z \sim p(\cdot | x, \kappa^{(t)}, \mu^{(t)}, \Sigma^{(t)})} (L(\kappa, \mu, \Sigma; x, z)) = \sum_{i=1}^n \bar{E}_{z \sim p(\cdot | x^{(t)}, \mu^{(t)}, \Sigma^{(t)})} (L(\kappa, \mu, \Sigma; x_i, z_i))$$

$$= \sum_{i=1}^n \sum_{j=1}^K (\underbrace{p_{ij}^{(t)}}_{\downarrow} (\log \kappa_j + \underbrace{\log \mathcal{N}(x_i; \mu_j, \Sigma_j)}_{\downarrow}))$$

notice that we can maximize them independently

first part: maximize λ

$$\arg\max_{\lambda_1, \dots, \lambda_k} \sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)} \ln \lambda_j$$

s.t. $0 \leq \lambda_j \leq 1 \rightarrow$ this is not necessary.

$$\lambda_1 + \dots + \lambda_k = 1$$

Since we can deal with logs $\Rightarrow \lambda_j > 0$

and $\lambda_1 + \dots + \lambda_k = 1$

$\Rightarrow \lambda_j \leq 1$. So the result will satisfy this automatically if exist

Lagrange form:

$$\mathcal{L}(\lambda_1, \dots, \lambda_k, \lambda)$$

$$= \sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)} \ln \lambda_j + \lambda (\lambda_1 + \dots + \lambda_k - 1)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_j} = \frac{\sum_{i=1}^n p_{ij}^{(t)}}{\lambda_j} + \lambda = 0$$

$$\Rightarrow \lambda_j = \frac{\sum_{i=1}^n p_{ij}^{(t)}}{-\lambda} = \frac{\sum_{i=1}^n p_{ij}^{(t)}}{\sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)}}$$

put $\lambda_1 + \dots + \lambda_k = 1$ in, so we have $\frac{\sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)}}{-\lambda} = 1$

Second part: maximize μ and Σ

$$\arg\max_{\mu_j, \Sigma_j} \sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)} \ln \mathcal{N}(x_i; \mu_j, \Sigma_j) \quad (2\lambda)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$$

$$\Leftrightarrow \arg\max_{\mu_j, \Sigma_j} \sum_{i=1}^n \sum_{j=1}^k p_{ij}^{(t)} \left(-\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\Sigma_j)) - \frac{1}{2} (x_i - \mu_j)^T \Sigma_j^{-1} (x_i - \mu_j) \right)$$

$$\frac{\partial}{\partial A} \ln(\det(A)) = (A^{-1})^T$$

$$\frac{\partial}{\partial x} (x^T A x) = (A + A^T)x$$

take derivative w.r.t μ_j

$$\Rightarrow \sum_{i=1}^n p_{ij}^{(t)} 2 \Sigma_j^{-1} (x_i - \mu_j) = 0 \quad \frac{\partial}{\partial A} (x^T A x) = x x^T$$

$$\Rightarrow \sum_{i=1}^n p_{ij}^{(t)} \Sigma_j^{-1} x_i = \sum_{i=1}^n p_{ij}^{(t)} \Sigma_j^{-1} \mu_j \quad \frac{\partial}{\partial A} (x^T A x) = -A$$

$$1A^{-1} = \frac{1}{1A}$$

$$\Rightarrow \mu_j = \frac{\sum_{i=1}^n p_{ij}^{(t)} x_i}{\sum_{i=1}^n p_{ij}^{(t)}}$$

let $\Lambda_j = \Sigma_j^{-1}$ precision matrix

$$p_{ij}^{(t)} \left(-\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\Lambda_j)) - \frac{1}{2} (x_i - \mu_j)^T \Lambda_j (x_i - \mu_j) \right)$$

$\Rightarrow$ take derivative w.r.t Λ_j

(if w.r.t Σ_j , will get the same result, but much harder)

$$\sum_{i=1}^n p_{ij}^{(t)} \left(\Sigma_j^{-1} - \frac{1}{2} (x_i - \mu_j) (x_i - \mu_j)^T \right) = 0$$

$$\Rightarrow \Sigma_j = \sum_{i=1}^n p_{ij}^{(t)} (x_i - \mu_j) (x_i - \mu_j)^T$$

$\rightarrow$ the variance collapse to 0 when $p_{ij}^{(t)} = 1$ for a i